

Unsupervised Value Discovery: Locating the Value Structures of an Agent from Passive Observation by their Empowerment–Plasticity Asymmetry

SJ Beard^{*1,2}

¹Principles of Intelligence

²Centre for the Study of Existential Risk, University of Cambridge

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Abstract

Unsupervised agent discovery (UAD) locates agents in raw multivariate time series by finding variable sets that approximately satisfy a Markov-blanket condition against their surroundings. We ask a question one level inward: within a discovered agent, which parts are its *value structures* — the components of its informational structure that correspond to its goals, objectives, desires or utility functions? We propose that this role has a directional information-theoretic signature: values are the components that most strongly drive the world while being least driven by it — maximally empowered and minimally plastic. This inverts a familiar assumption due to Omohundro that a system first has values and then acts to make them empowered and stable; we hypothesise instead that whatever is empowered and stable thereby *serves as* the system’s values, and that this structural property is passively detectable. We operationalise the hypothesis as a *value signature* on directed-information scores gated by procedure-mirroring noise floors, and evaluate it in three pre-registered experiments on planted-structure worlds built on an existing UAD benchmark. (i) In a single-agent world, the signature recovers the planted goal, and only it, in every measurement convention able to see through the world’s XOR-style composition; a naive pairwise convention fails structurally. (ii) Across four worlds including an eight-agent, 49-variable colony, only a block-level convention that fuses discovered agents and offers every block as a conditioning “key” recovers every planted goal; all variable-level conventions hit a scale wall, caught by a control. (iii) Against adversarial impostors, a fast environment-driven “captured goal” is refused, but a slow, history-triggered captured goal defeats the signature — traceable, a follow-up comparison showed, to the signature’s leak-tolerant intake clause rather than to any limit of passive observation: a strictly gated passive stability test refuses the same impostor — and a noiseless goal twin is passively indistinguishable in principle; an interventional yardstick generalising the benchmark’s goal-flip operation resolves both cases, and the twin case admits no passive resolution, locating the premium of interventional access precisely. We report limitations (single-variable values; lag-1 conditioning) and the directions they motivate.

1 Introduction

The standard route to an agent’s objectives runs through its behaviour: observe choices, infer the utility or reward that rationalises them [4, 6]. This route has well-known difficulties — non-

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identifiability, dependence on rationality assumptions — and one that is rarely foregrounded: an agent may *conceal* its objectives, for instance by injecting noise into its decisions, so that the behavioural signal is deliberately weakened. This paper explores a complementary route. Rather than inferring values from what a system does, we look for its value structures in the *informational structure* of the system itself: which internal component most drives behaviour beneath the noise? If such a component can be located structurally, it can be isolated and studied directly, recovering the signal that behavioural inference alone would lose.

Our starting point is unsupervised agent discovery (UAD) [8], which finds agents in raw time series as variable clusters that approximately satisfy a Markov-blanket condition against their environment [2, 3]. UAD locates agents’ *boundaries*. We ask what *drives* them, and specifically what plays the role of a discovered agent’s values. We deliberately say *value structure* rather than *value function*: locating the component that plays the value role does not tell us what that component does or prefers, and we have not built tools for that.

The hypothesis: inverting the classic AI drives argument. Omohundro’s *Basic AI Drives* [5] argues that an advanced system’s utility function “will be precious” to it, so the system will act to make that function maximally influential and maximally protected from change — in the vocabulary of Abel et al. [1], maximally *empowered* and minimally *plastic*. Omohundro’s direction of explanation runs from values to structure. We hypothesise the reverse: whatever component of a system is maximally empowered and minimally plastic will, by that fact, *serve as* the system’s values. This is why we speak of *value* discovery rather than *goal* discovery: we are identifying the parts that play the value role, whatever they are. In the simple systems studied here that part is a single planted goal bit; the same structural signature could in principle locate richer value structures in richer systems.

The empowerment–plasticity trade-off [1] makes the hypothesis non-trivial: across a boundary a system cannot be maximally both, so an adaptive system has reason to specialise internally — belief-like parts skewed towards plasticity, value-like parts towards empowerment, decision-making between. Our claim is that this specialisation is passively detectable.

Contributions. (1) A pre-registered operationalisation — the *value signature* — with procedure-mirroring noise floors and controls, designed so a non-technical collaborator could audit every step in plain English. (2) A benchmark of measurement conventions across four worlds identifying a block-level convention (fuse discovered agents; offer every block as a conditioning key) that recovers every planted value at a scale where all variable-level conventions fail. (3) A characterisation of the instrument’s limits by adversarial impostors, together with an interventional yardstick that generalises the UAD benchmark’s goal-flip operation and resolves both defeat cases — one of which (the zero-noise twin) is provably beyond any passive method, while the other, a later comparison showed, is not (see the Discussion). (4) Several mechanism-level findings (an XOR “cipher” that hides inputs from under-conditioned measures; the self-penalising property of procedure-mirroring floors; a “twin channel” by which a disconnected value persists in the signature through its own record).

2 The value signature

2.1 Directed measures

Let X be a candidate variable in a multivariate discrete time series and R “the rest” under some convention. Using the smoothed plug-in conditional mutual information of the UAD implementation

(Laplace $\alpha = 0.1$, nats), we define lag-1 directed flows of transfer-entropy type [7]:

$$\text{intake (push-in): } I_{\text{in}}(X) = I(X_{t+1}; R_t | X_t), \quad (1)$$

$$\text{drive (push-out): } I_{\text{out}}(X) = I(R_{t+1}; X_t | R_t). \quad (2)$$

The *polarity* $(I_{\text{out}} - I_{\text{in}})/(I_{\text{out}} + I_{\text{in}}) \in [-1, 1]$ is the fraction of a variable’s traffic that is outbound; a raw difference is always reported alongside. Different conventions instantiate R differently (Section 3.1); an environment-flavoured drive toward a designated environment variable is also computed, never conditioned through the agent’s action, because the value’s path to the world runs through the action and conditioning on a mediator screens off precisely the flow of interest (verified analytically in Section 4).

2.2 Floors

Every score is compared to its own null distribution: the candidate’s column is circularly shifted against the fixed remainder (≥ 200 shifts, pooled across seeds where registered), and the *identical* scoring procedure is re-run — including any maximisation over conditioning sets, so that search-induced bias is priced into the floor. A score counts as *above floor* if it exceeds the 95th null percentile *and* clears the null mean by $z \geq 3$ null standard deviations (the margin gate was added after a near-frozen variable was found to pass hair-thin residuals; see Section 5). We write $\text{net}(\cdot) = \max(0, \text{score} - \text{floor}_{95})$.

2.3 The signature

Definition 1 (Value signature). *A variable X exhibits the value signature if (i) its drive is above floor, (ii) its total flow $I_{\text{in}} + I_{\text{out}}$ is above floor (it is rankable), and (iii) $\text{net}(I_{\text{out}}) > 0$ and $\text{net}(I_{\text{out}}) \geq 9 \text{net}(I_{\text{in}})$.*

Clause (iii) is deliberately leak-tolerant: real components are all affected by their environment, so a value must not be disqualified for measurable intake — only for failing to be overwhelmingly a driver (the $9\times$ ratio corresponds to polarity ≥ 0.8 and sits well above the $\approx 3\text{--}4\times$ observed for genuine mediators). “Intake at floor” is the special case $\text{net}(I_{\text{in}}) = 0$.

3 Experimental design

Worlds. All worlds derive from the `SyntheticHandleWorld` of the UAD handle benchmark [8]: nine binary variables per agent forming the loop belief $B \rightarrow \text{action } A \rightarrow \text{environment } E \rightarrow \text{sensor } S \rightarrow B$, with a goal G entering the policy ($A = B \oplus G$ up to noise) and flipping by its own coin at rate 0.015; plus a distractor D shadowing E , pure noise W , and in the original world two perfect “lookalike” logs of the sensor and action lines. The world was built by its author before this project existed and for a different purpose, so it could not have been tuned to flatter our measure. Table 1 lists the worlds; each of ours ships with an audit — change-frequency chart, a mechanism table comparing observed disagreement rates against those predicted from the designed noise (agreement to $\lesssim 0.1\%$), and a plain-language “story” — reviewed before any scoring.

Conditions and controls. Each world runs in three conditions: *main*; *no-core* (goals present but disconnected from the policy — nothing may exhibit the signature); and *scramble* (every column independently time-shuffled — the count of above-floor scores must sit in the 5% chance band, which doubles as a calibration check on the floors). A contrast world from the UAD authors’ older

Table 1: Worlds. All ring-coupled colonies have eight agents (≈ 49 variables) with a neighbour’s action nudging each agent’s patch on $\approx 10\%$ of steps.

World	Variables	Planted structure / probe
Anchor (v1)	9	one goal at 1.5%; two perfect lookalike logs
Colony (v2)	49	eight goals; ring coupling; partitioned variant for attribution
Deep-synergy (v2)	17	two agents, two beliefs each; parity vs. majority rule
Slow-meter (v2)	14	goal slowed to 0.5%; four causally inert meters
Puppet colonies A1/A2 (v3)	49	one goal replaced by a <i>captured</i> goal (fast / slow-triggered)
Alias colony (v3)	51	goal twins for two agents at noise $\{0, 0.5, 1, 2, 5\}\%$

Table 2: Measurement conventions. “Own-block” conditions on the candidate’s own agent block; “any-block” offers every block (and singleton) as a conditioning key.

Convention	Level	Rest / conditioning
pairwise	variable	sum of one-at-a-time flows
fused	variable	joint state of all other variables, both directions
fused + best-key	variable	fused intake; drive = best over no key and each single key
grown keys	variable	fused intake; drive = greedy key growth (cap 2)
fused-agents own-block	block	own agent block standing in the conditioning
fused-agents any-block	block	target block’s past + best single key element

multi-agent simulator supplies a variable literally named `goal_progress` — a causally inert progress meter — that the instrument must refuse.

Pre-registration and discipline. Success criteria (thresholds $\geq 80\%$ of 20 seeds; $\geq 90\%$ for goal-above-belief; uniqueness = no non-goal variable holding the signature in more than 2 of 20 seeds) were locked before each sweep; every calibration amendment made before a sweep is dated in a decisions log. Two working rules shaped the analysis: when two requested measures disagree, the disagreement is investigated and mechanistically explained rather than averaged; and conventions were run in parallel as co-equal tests rather than a primary being chosen early. Ground-truth labels were used only to evaluate outcomes, never inside any measurement.

3.1 Measurement conventions

Table 2 lists the conventions (*tests*) evaluated. The variable-level conventions differ in how R is instantiated; the block-level conventions first run the UAD detector to partition variables into agents (its cluster-count dial swept, the dial chosen by blanket-valid coverage as in the UAD authors’ own selection procedure), fuse each agent’s variables into a compressed macro-state block (top-64 states; lost mass reported), and never fuse the environment — agent blocks respect a *discovered* coherence, the environment has none to respect. Orphaned variables (those the detector drops — characteristically the goals, whose rare flips carry almost no symmetric mutual information) remain individually scoreable.

Table 3: Experiment 1 (anchor, 20 seeds). Rates are fractions of seeds. Uniqueness = nothing but G holds the signature. C3 = `goal_progress` refused.

Convention	G signature	unique	$G > B$	controls	C3 refused
fused	1.00	0.95 / 1.00	1.00	pass	—
fused + best-key	1.00	0.95	1.00	pass	0.70
pairwise	0.05	0.00	0.05	pass	1.00

4 Experiment 1: does the signature recover a planted value?

The anchor world was scored under pairwise, fused, and fused+best-key, in the world as-is and with the lookalike columns removed, on 20 seeds. Before any scoring, the estimator was verified against analytic targets (a delayed copy reads $\ln 2$ nats; a chain $X \rightarrow Y \rightarrow Z$ shows $I(Z_{t+1}; X_t | Y_t) \approx 0$ while the unconditioned lag-2 dependence is positive — the mediation exhibit; the Marko–Massey conservation identity holds on analytic cases).

The cipher. A first pass showed the action’s intake reading ≈ 0.005 nats pairwise while a pre-planned fused cross-check read 0.31. Investigation, prompted by the disagreement rule, revealed the mechanism: XOR composition ($A = B \oplus G$) is a perfect cipher — each input is invisible to a measure conditioning on too little unless the other input (the “key”) is conditioned on. Measured readings matched hand-derived values to three decimals; re-timing the world did not help (tested), and a single extra conditioning variable restored the goal’s drive on the environment from 0.000 to 0.441 nats (hand-predicted 0.446). This motivated the key-based conventions and, later, the block-level ones.

Results. Table 3 summarises. Under fused and fused+best-key the goal exhibits the signature in 100% of seeds, uniquely in $\geq 95\%$, and lies above the belief in polarity in 100%; all controls pass; the only intruder anywhere was pure noise W in one seed in twenty — the false-positive rate the floors are designed to admit. Under pairwise the action wires share the signature in 20/20 seeds (as pre-registered) and, with honest floors, the goal’s own pairwise signal is indistinguishable from nothing: pairwise measurement cannot tell a value from a wire in a world with encrypted composition, and was retired with its cause of death documented. Removing the lookalike columns changed no verdict, so raw data needs no pre-cleaning. The UAD detector run on the same trace recovers the agent’s body as one blanket-valid cluster (leakage 0.001 against tolerance 1.0) but drops the goal as an isolated singleton: boundary-finding and value-finding are complementary axes.

The C3 residual and the floor standard. Best-key refused `goal_progress` in only 70% of seeds. Reading the stored records falsified our first mechanism story: the thief was a *near-frozen* column (change rate ≈ 0.000) whose every score sat at its own unstable floor, plus initial transients from an un-burned-in simulator. This produced the z -margin gate, a burn-in rule, and a reproducibility fix for the older simulator (its traces are cross-process reproducible only with a pinned hash seed). A residual 30% theft by a genuinely moving meter is real Granger flow with no causation — diffuse shared-history witnessing that no single conditioning key can screen — and became a benchmark requirement for the conventions of Experiment 2.

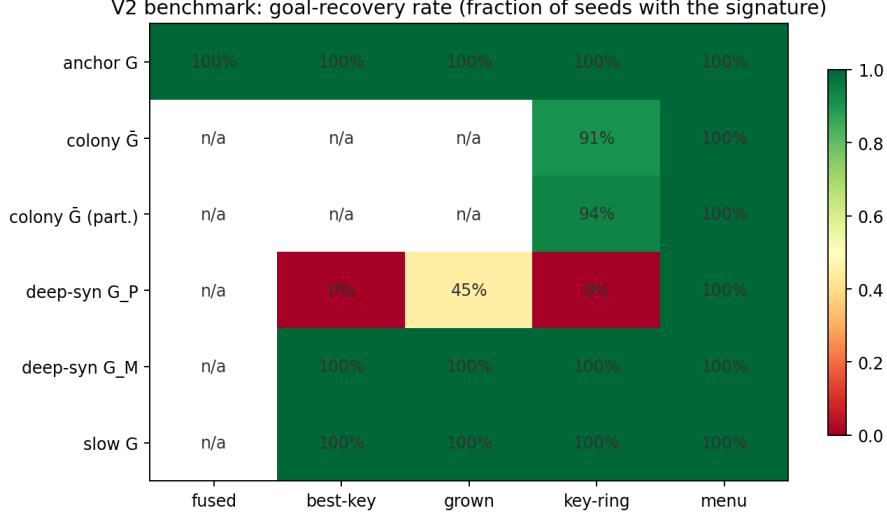


Figure 1: Experiment 2: fraction of seeds in which each planted goal exhibits the value signature, per convention (columns) and world (rows). “n/a” marks recorded infeasibility (fused beyond ~ 10 variables; best-key/grown intake at colony scale). Any-block is the only column without a miss.

5 Experiment 2: which conventions survive more complex worlds?

Two new conventions were qualified against the full anchor criteria before entering the benchmark. *Grown keys* exposed its own limit during its justifying unit test: greedy growth cannot climb a flat plateau — in pure three-way parity every single key has zero marginal gain — while excelling on graded composition (majority rules, +43% over best-key); it passed C3 at 100%, falsifying our prediction that it would inherit best-key’s witness blindness, because *procedure-mirroring floors self-penalise search depth*: a deeper maximisation must find real structure to outrun its own inflated null. The two block-level conventions also qualified (100% on every criterion including C3; the block context absorbed even the chance blips seen elsewhere).

Fig. 1 shows goal-recovery rates across the four worlds. **Any-block recovered every planted goal in every world** — the only convention that never missed. Mechanistically, “target block’s past plus one key element” is two-block conditioning, which decrypts three-way composition and, crucially, does not require the goal to belong to a detected agent: since the detector orphans goals, this robustness is fitness for the terrain. Own-block, conditioning only on the candidate’s own block, has no ring for an orphaned goal (0% on the parity goal; 70% worst-goal in the coupled colony); the divergence between the two is now our partition-quality diagnostic. Every variable-level convention hit a scale wall: fused’s is combinatorial (state count doubles per variable), and best-key/grown’s fused-style *intake* saturates outright at 49 variables (19,999 distinct joint states in 20,000 samples), so intake reads as bias for every variable — a defect the *no-core control caught* (0% clean seeds), which the locked interpretation rules converted into a recorded infeasibility rather than a spurious result. The synergy cliff was measured on the parity goal (best-key 0%, grown 45%, any-block 100%; the majority goal 100% for all), and a pre-registered contingency — re-running colony failures with coupling removed — attributed both own-block’s shortfall (70% $\rightarrow$ 90%) and the single systematic thief (an action wire, 6/20 $\rightarrow$ 2/20) to ring interference, not scale.

Two defects of our own were found and fixed mid-programme by the same discipline: an early deep-synergy world drove both belief channels from one action, so their XOR was quasi-frozen and

the parity silently collapsed (the audit checked marginals only; it now checks joint dynamics), and the feasibility map had missed the intake wall.

6 Experiment 3: how easily can the instrument be fooled?

Experiments 1–2 planted values and asked whether they could be found; Experiment 3 planted impostors and asked whether they could fool the surviving conventions (any-block primary, own-block diagnostic), in colony-style worlds, with an interventional yardstick alongside.

The yardstick. The UAD benchmark embeds a `goal_flip` operation — a *handle*, not an ideal intervention: applied to the true goal it changes the action within the tick; applied to anything else it does nothing. We generalised it to a *do-style flip* of any variable’s value as consumed by the dynamics, on 50% of interleaved steps, and built the two-direction *interventional mirror* of the signature: a *goal-role probe* (does flipping X change its agent’s behaviour, given the agent’s policy context?) and an *autonomy probe* (does flipping X ’s own environment change X ?). The interventional verdict — role effect above floor *and* autonomy effect at floor: “levers the world, unlevered by the world” — is the signature restated under access. Its construction was itself instructive: the XOR cipher bit three times (an unconditioned role probe; a saturating autonomy readout; unconditioned autonomy terms) and was caught each time by a validation bench on the anchor, where the do-style flip had to agree with the embedded operation before use; the fix, each time, was the context conditioning the benchmark’s own interventional score already employs.

Results. Table 4 gives the access-premium table. The fast captured goal — wired into the action but copying its own patch — was refused in 20/20 seeds (its intake is large and visible; the audit also recorded an emergent effect: copying the patch its agent’s belief copies, it partially cancels the belief and *pacifies its own environment*). The slow captured goal — flipping at a true goal’s tempo (1.77%), every flip verified to follow a six-step run of its patch — **earned the value signature in 20/20 seeds under both passive tests**, exactly as pre-registered. Our reading at the time — that one step of conditioning cannot see a trigger living in run history — turned out to be too strong. The impostor’s environment-intake was in fact above its noise floor at lag 1 (net ≈ 0.03 nats), but its drive-to-intake ratio ($\approx 14\times$) passed the signature’s leak-tolerant clause (§2), which was adopted so that real, leaky value cores are not disqualified; a subsequent comparison experiment (v5; Discussion) showed that a strictly gated passive stability test refuses this impostor at lag 1. Within the signature family, multi-horizon conditioning remains the evidenced remedy (the intake also surfaces above the lag-1 floor at lag 2 in 20/20 seeds). The yardstick unmasked it in 20/20 (role yes, autonomy yes). Fig. 2 gives the passive breaking point for the goal role: twins are indistinguishable from their goals at 0–0.5% noise (as they must be — identical sequences cannot be separated by any passive method), transitional at 1%, and separated at $\geq 2\%$ (19–20/20), with no inversions; the yardstick refuses the bit-identical twin at 0% (a recording levers nothing) while its goal keeps the verdict. Interventional access resolves both defeat cases; only the zero-noise twin is, after the v5 comparison, known to *require* it.

The twin channel. In the alias no-core condition the *disconnected* goals still exhibited the signature in 20/20 seeds — correctly: a goal stripped of its behavioural wiring still causally drives its own recording, so “drives without being driven” genuinely holds through the record. The instrument was right and the control’s expectation was mis-specified for twin-bearing worlds. Philosophically the point is sharp: a value can persist in the signature through its records alone.

Table 4: Experiment 3: the access-premium table (20 seeds; passive = any-block).

Impostor	Best passive verdict	Yardstick verdict	Premium
Fast captured goal	refused (20/20)	refused (20/20)	none
Slow captured goal	signs (20/20)	refused (20/20)	total
Twin, 0–0.5% noise	indistinguishable (19–20/20)	twin refused (20/20)	total
Twin, $\geq 2\%$ noise	separated (19–20/20)	separated	none

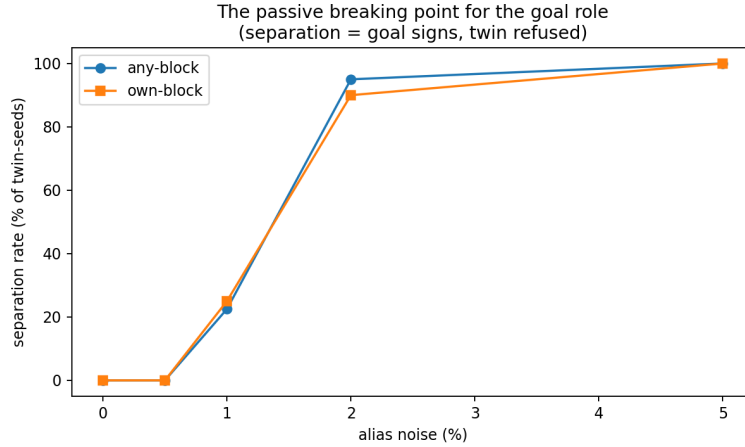


Figure 2: Experiment 3: separation rate (goal exhibits the signature, twin does not) versus twin noise — the passive breaking point for the goal role lies between 1% and 2%.

7 Discussion

What was established. The empowerment–plasticity signature locates a planted value structure from passive observation, uniquely, across single- and multi-agent worlds, and refuses decoys, disconnected goals, scrambles and (with the right conditioning) meters; the block-level any-block convention is the only tested instrument whole at the scale of the UAD authors’ multi-agent simulator. Its limits are now measured rather than conjectured — history-triggered captured goals defeat the signature’s leak-tolerant intake clause (not, as we first wrote, lag-1 passivity itself — their intake leak is measurable and a stricter passive gate refuses them); twins below $\approx 1\text{--}2\%$ noise are passively indistinguishable — and interventional access closes both gaps, with the twin gap admitting no passive closure, giving a quantitative form of the access-model thesis of Zarncke [8]. Several findings concern method rather than result: the XOR cipher; the self-penalising floors; the collapse-detector role of the no-core control; and the value of running disagreeing conventions in parallel rather than picking a primary early — twice, the convention we would have discarded first turned out to carry the mechanism.

Why structure, not behaviour. Behavioural inference and structural discovery are complementary, but structure has a distinctive advantage under adversarial conditions: an agent that conceals its objectives by adding decision noise weakens the behavioural signal, yet the component that most drives behaviour beneath that noise remains structurally locatable; once located, it can be isolated and studied in a regime where the noise no longer obscures it. Our results are a first indication that such structural location is possible.

Limitations. (1) *Single-variable values.* The instrument scores individual variables. It has been tested on agents with multiple belief states but not with multiple goal or value states; a value structure distributed over several components is beyond it as it stands. The grown-keys results suggest one route (deciphering interactions among several value-like variables); we conjecture a better one — use UAD’s boundary-detection machinery to identify sub-systems within an agent, then evaluate the empowerment and plasticity of those sub-systems rather than of single variables. (2) *Lag-1 conditioning together with the leak-tolerant intake clause,* which the slow captured goal jointly exploits; multi-horizon conditioning and a stricter stability gate are both evidenced remedies (the latter from the v5 comparison) — what strictness costs on genuinely leaky value cores is untested. (3) *Planted structure.* All worlds are engineered; whether the polarisation *emerges* in evolved or learned systems is the question the instrument now makes askable, not one it has answered. (4) The protocol was soft-blind: the implementing system had read the world’s source; protection came from the world’s independent origin, pre-registration, controls and multi-seed sweeps. (5) The relation between this signature and Omohundro’s drives is a hypothesis supported by recoverability, not a demonstration that empowered-and-stable components *function* as values in richer agents.

Subsequent experiments (v4 and v5). Two further experiments post-date the work reported above and inform the corrections woven into this text. *v4* pointed the instrument at an event-log deployment simulator carrying an injected self-preserving utility: the value’s structural footprint there is the drive-without-intake of the *actions the utility selects for* (regime-graded, and visible in the auditor’s log alone), not the outcome state the utility rewards — in event-log worlds the mark of a value is on the choices, not the scoreboard.¹ *v5* ran the signature side by side with a role-sufficiency formalism independently derived by another PIBBSS mentor (roles as minimal internal representations sufficient for named external targets; interventional at full strength).² The comparison converged wherever the formalisms overlap — including an interventional battery identical to our yardstick — and diverged instructively twice, in mirror image: a strict stability gate (theirs) passively unmask the slow captured goal our leak-tolerant clause admits, while an evidence-only stability quantifier (theirs) lets the witness class steal the goal verdict our all-source intake refuses. A new world with the value structure *outside* the agent it drives was found by the signature in 20/20 seeds while remaining invisible to any internal-only candidate scope; and on observation-only data the sufficiency tests cannot be stated at all, while the passive signature runs. The active-vs-passive contrast — what needs a do-operator, what passivity recovers, and what can even be asked of data that only watches — is, on this evidence, the load-bearing difference between the two routes to the same construct.

Future work. Systems-of-agents environments in which values may be held by coalitions rather than individuals (any-block requires no assumption about which agent owns a value, and so is structurally open to this); multi-horizon conditioning; sub-system-level scoring for distributed value structures; the twin channel as a probe of value persistence through records; and a synthesis instrument combining the signature’s world-directed, all-source intake with the strict stability gate and an explicit uniqueness pressure (the v5 reversals define its requirements).

¹<https://github.com/SJ-Beard/deployment-pipeline-value-detect>

²<https://github.com/SJ-Beard/active-v-passive-value-detect>

Statement on use of AI

The author conceived the hypothesis and framing, made all design decisions, reviewed every audit and pre-registration in plain language, and directs the programme. Claude (Anthropic), an AI system, wrote all code, ran the experiments and analyses, and drafted this manuscript under the author’s direction; every decision, deviation and defect is dated in the project’s decisions log. **This manuscript is, as it stands, AI-generated:** it was written by Claude and reviewed and approved by the author, who intends to write the account in their own words when time allows. In line with the standing convention that AI systems are not listed as authors — authorship carries a responsibility that cannot be delegated to them — the author takes full responsibility for the content. The manuscript is not being submitted for publication.

Reproducibility

All code (`value_detect`), world generators, audits, locked pre-registrations, verdict tables and figures are archived with fixed seeds; the UAD benchmark code is imported unmodified. Sweeps ran on a single laptop ($v1 \approx 1$ h; $v2 \approx 12.5$ h; $v3 \approx 37.5$ h at four workers).

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